[G0R72A] Data Visualisation Project Report "Exploration"

Part 1. Metadata

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- Group number: ingroup_5
- Dataset: oceanus
- User: Elena Petrova

Part 2. Project description

The dataset centers on a governmental oversight board - Commission on Overseeing the Economic Future of Oceanus (COOTEFOO), which aims to improve the economy of the city of Oceanus. The city of Oceanus nevertheless is currently divided into two main groups: ones who want the board to keep focusing on the legacy fishing-based industry, which is represented by the group Fishing is Living and Heritage (FILAH) and the ones who prefer the board to hedge their support to the fledging tourism industry, which is represented by the group (Tourism Raises OceanUs Together (TROUT)). Both of these groups accuse the board of being biased against their interest and for the other.

The data set contains 2 data sources, which have identical structures: one provided by the board themselves, and one provided by an investigative journalist, which contains more data points. The data provided contains the following main information:

- **Topics**: The topics that the boards and its members are participating in to improve Oceanus' economy. We can sort out these topics into pro-fishing, pro-tourism and vanity/neutral as below (except for seafood festival which benefits both):

Pro-fishing	Pro-tourism	Neutral
- Deep maintenance to the commercial fishing dock at Himark - Money for a new crane at Lomark - Fish vacuum - Low-volume unload crane in Haacklee - Affordable housing for fishing workers	- Expanding the tourist wharf /infrastructure in Haacklee - Statue to honor John Smoth in Port Grove - Establishing a Marine Life Observation Deck in Centralia - Developing a Heritage Walking Tour in Haacklee - Establishing a Waterfront Market in Haacklee - Plans for a music concert	- Renaming the park. - Putting a name on pictureque harbor area in Port Grove - Putting a name on the inspection office in Lomark

- **Discussions, Meetings, and Plans:** Discussions, meetings, and plans that are linked to the above topic. These meetings and discussions, meetings, and plans can be attended by one or multiple board members, of which their **sentiments** are recorded by a value of -1 to 1 towards the topic; -1 means fully disagree and 1 means fully agreed.
- **Trips:** The lists of trips the board members participated in as well as the places they visited and the start and the end of their trips. For each trip, there is only 1 board member going.
- **Places:** Places in Oceanus can be linked by the travel plans or referred to in the discussions; they also have coordinates that can be used in a map of Oceanus (github.com/vast-challenge/2025-data/blob/main/MC2_release.zip/oceanus_map.geojson).

The persona we are supporting is Elena Petrova, a member of FILAH, which aims to find clues that show whether fishing is being neglected, misrepresented, or treated unfairly compared with tourism. Some directions that we looked into are the following:

- Whether there are sentimental biases of the board members towards tourism projects compared to fishing projects?
- Do board members spend more time on tourism projects compared to projects that benefit fishing?
- Whether the board attempted to remove these biases / preferences when publishing their data (compared to the uncensored journalist data)?

Part 3. Visual design

To create the final designs, we performed a collaborative process based the Diverge – Emerge – Converge process. To begin with, we perform on our own personal analyses and create own own individual sketches. In order for each other to avoid each other ideas, we uploaded our graphs to Miro and to our WhatsApp group during the week. Additionally, we also looked at Toledo recordings as inspiration for our graphs and look at the original source of the dataset (VAST Challenge 2025 MC2), where we additionally got the map file of Oceanus.

After a week, we met in order to discuss both the emergence and convergence of our initial graphs in 2 sessions. In the first session, we focused mostly on the Emerge part, where we explained our graphs and criticized each other. Then, we modified the graphs based on the SCAMPER method. We made multiple iterations of our graphs, with an aim to make them more visually appealing as well as being more 'integratable" to each other's designs. Additionally, we have some additional brainstorming time to get more initial graphs to add into the Emerge process.

Finally, in the second in-person session "Converge", we focused on finalizing our small graphs and integrate them into big interactive visualizations that can answer the questions that was mentioned in Part 2. In more details, we discuss the connectivity or interactivity of our graphs and how helpful these graphs for our persona are in combination.

Cluster 1: Board sentiment biases, participation and concealed activity/biases

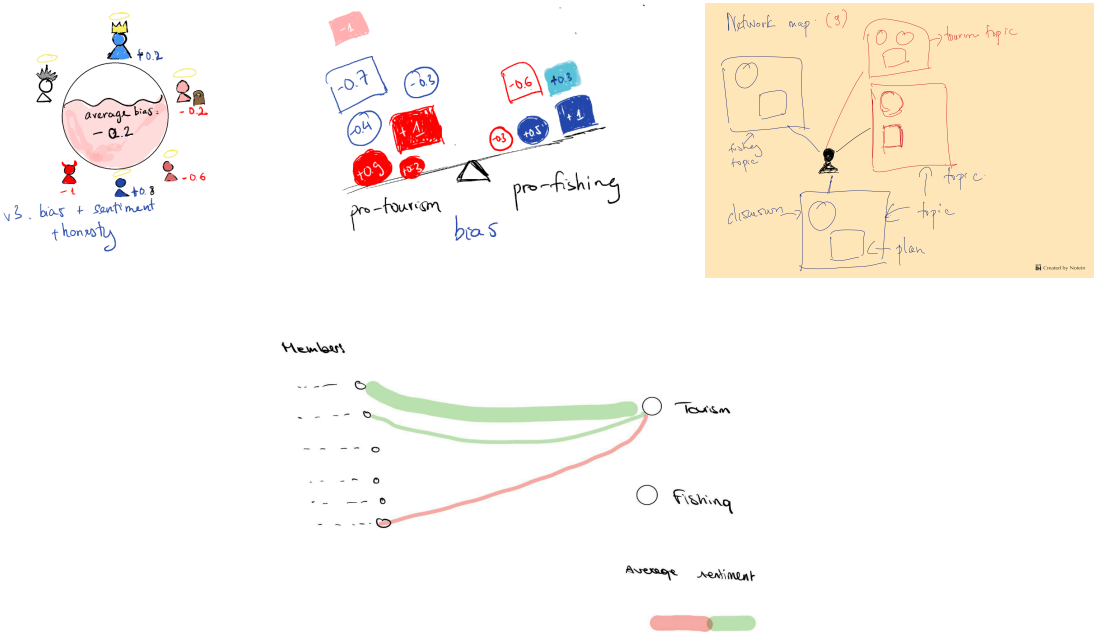


Figure: left to right, top to bottom: sketches C1, C2, C3 and C4

The first sketch, **C1**, aims to represent the **average sentiment** of the COOTEFOO board and its members. The position and design of each human glyph are set so that the user can easily visualize the role and authority of each board member: members with higher positions are placed above others and are also given crowns to indicate their exact role (a king's crown for the chairman, a queen's crown for the vice-chair, and a chest for the treasurer). The average sentiment is depicted by the color of the human

glyph as a gradient: the redder the color, the more pro-tourism and anti-fishing the person is, and conversely, the bluer the color, the more pro-fishing the person is. At the center, the average sentiment of the board is denoted with both a numeric value and a wave fill. Lastly, the halo/devil-horn glyph indicates whether the person is publishing all of their sentiments in their government data or not.

The second sketch, **C2**, aims to show the **detailed sentiments** of a person (or the whole board) for each discussion and plan. The sentiments are placed on a scale, with pro-tourism on one side and pro-fishing on the other side. The shape represents the type: discussions are circles and plans are squares. Similar to C1, colors are used and are filled if the person agrees with the discussion or plan; they are marked blue or red depending on the topic to which they belong. The opacity of the shape indicates whether the item is published by the government or not: items that appear only in the journalist's data are slightly transparent.

The third sketch, **C3**, is a **relationship map between the user and the topics** and their subitems. The topics are color-coded blue or red depending on whether they support fishing or tourism. While some of these items are similar to those in C2, this visualization provides a more holistic view of the number of fishing-related and tourism-related topics.

The fourth sketch, **C4**, is another representation of the average sentiment. In this design, a thicker line between a person and fishing/tourism indicates a stronger positive or negative view, while the colors green and red indicate the direction (positive or negative).

By combining **C1, C2, C3, and C4**, we obtain the clustered design displayed below. The C1 chart is kept as is and used as the center of the new visualization. We then link the human glyphs from C3 to C1 to display participation in detail for each discussion and topic. The links between board members and discussions or plans are replaced with the variable lines from C4, where the thickness represents the absolute sentiment and the color indicates the direction. We also color-code the average sentiment for each discussion and plan similarly to C2. Additionally, to show concealed discussions and plans, the borders of these items, as well as any connections between them and the board members, are represented with dotted lines.

As a result, the combined visualization explains the sentiments of the board and its members not only toward fishing in general but also toward detailed activities. It also shows their participation and indicates whether the information was officially published or discovered by the journalist.

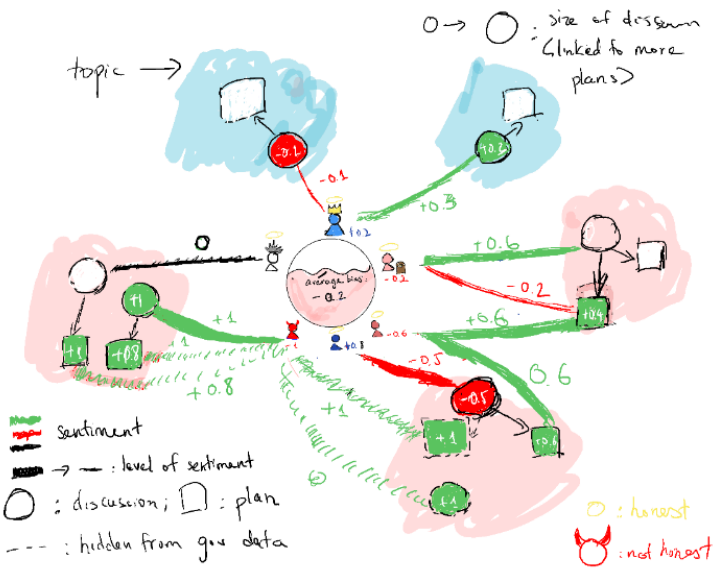


Figure: Converged cluster 1

Cluster 2: Board visit map and time spent over

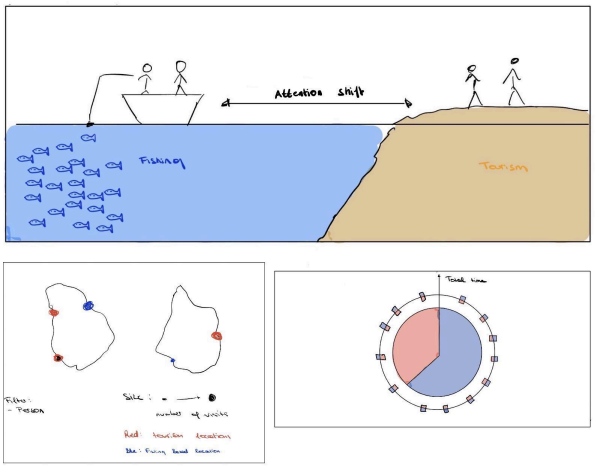


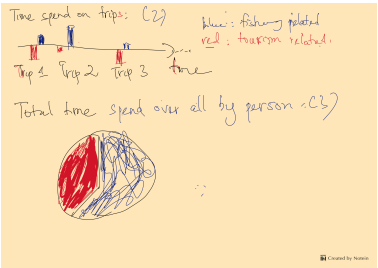
Figure: from left to right, top to bottoms: sketches C5, C6, C7 in cluster 2

- The first sketch, **C5**, represents the **fishing-tourism attention** (by time or number of visited places) of all board members. In detail, board members that have spent more attention to fishing finding themselves in the water and the board members that focused more on tourism being on the shoreline. The arrow represents the direction of stronger attention: the farther from the shoreline to the land, the more biased against fishing that board person is.
This shoreline metaphor that we use turns this balance between tourism and fishing into a clear visual that is easy to comprehend. The drawing helps to quickly tell whether attention is balanced across the board or whether it is more skewed towards either side.

- The second sketch, **C6**, shows the **board visit map**, to view where the visited destinations are plotted on the map, the red locations would represent the tourist-related visits, while the blue locations would represents fishing-related visits. The size of each point would then correspond to the number of visits made to that location. Additionally, there is a filter that allows view a single board member or compare multiple board members.

This visit map shows a spatial view of where the board members' attention is mostly focused, instead of only looking at what they say, it shows what location they actually visit and how often. The color contrast is there to distinguish between fishing and tourism locations and the cluster the size of the circle shows the repeated attention a specific place. Hereby we can discover whether visits are distributed evenly or whether board members tend to concentrate on one specific type of location.

- The third sketch, **trip time summary** (C7), has an inner circle that is split to show how the total trips' time is divided between fishing-related visits and tourism-related visits. The outer circle is a line showing the time spent on places related to tourism (red) or fishing (blue), each line representing each trip in clockwise order. This was created by joining 2 smaller subgraph C7.1 and C7.2 as below:



This visualization is a good summary of how the time of the board members is allocated across visits. The inner part shows the division between fishing and tourism total time spent on these activities, while the outer part of the circle shows how that total is built up across separate trips. This helps in distinguishing whether the board member does many short trips and fewer longer trips as well as consistency or change in preference / biases over time.

As an interactive cluster, these three visuals provide a more complete picture of each board member's behavior. Selecting a board member in the first graph automatically updates and pops up the other two visuals, so the viewer can directly see the member's visits and how their time is distributed across trips. By linking the three visuals, the cluster allows us to compare what a member appears to support with what they actually

spend time on. In this way, the interaction is useful for interpretation, because it helps build a more thoughtful understanding of whether a board member's behavior is consistent across different types of evidence.

Part 4. Implementation

In this section, you describe 3 of the implementations that you created to answer the questions in the brief. Limit yourself to three; if you want you can include additional ones in the screencast. These implementations may be of designs that come directly from part 3 (diverge, emerge or converge), or of new designs. For each visualisation, explain the visual encoding and the interactions: how to interpret the marks and channels in the chart. Include screenshots (with annotations if relevant). You do not have to describe additional plots that you made in e.g. R or Excel to answer the questions; this is about the custom visualisations.

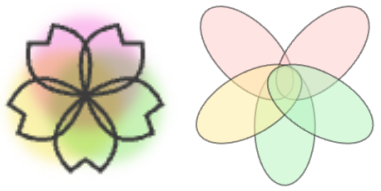
In the description below, we make a distinction between the intended/ideal design, and the actual design. The ideal design is what you want to have in the end (if you had the help from an experienced programmer); the actual design is what you ended up with. See figure below.

*Please provide the following **for each custom implemented visual**:*

1. *The intended design that you wanted to implement: a picture (hand drawn or a mock-up) of the design that you aimed for with the implementation. These could be from the diverge-emerge-converge stage earlier, or new ones. Important: they should be relevant for the questions mentioned in Section 2 of the report.*
2. *The actual design that you were able to implement: a description of the visual encoding of the implemented visual (what does each part refer to; what are the marks and channels). An annotated image/screenshot may be helpful.*
3. *A description of the interactions that are possible.*
4. *If the implementation does not have all features of the drawn sketch (in other words: if the actual implementation is not the same as the ideal one): how does your implementation relate to the target design? What is not included yet, or what needs to be added still if you had access to a skilled programmer.*
5. *A link to a running marimo instance, e.g. on huggingface (see https://docs.marimo.io/guides/deploying/#deploy).*
6. *A link to a 4-6 minute youtube video in which you (a) demonstrate the functionality of the custom visualisations you created, and (b) how this helps in answering the questions. Paste the full URL in the document instead of a link that automatically opens the web browser (i.e., https://www.youtube.com/watch?v=DLzxrzFCyOs instead of our_video). For instructions on how to upload a video, see e.g. https://www.youtube.com/watch?v=VtF2AgFSLAw. Important: set visibility to either "public" or "unlisted" (not "private").*

   Note: It is not forbidden to edit the video. You may include a title screen, but in that case make sure to not loose too much of your limited time doing that, because it is not the primary content.

Intended design (left) vs actual implementation (right). Taken from shirleywu.studio/filmflowers



Max 5 pages.

Implementation 1: Board sentiment biases, participation and concealed activity/biases

Description of the visual

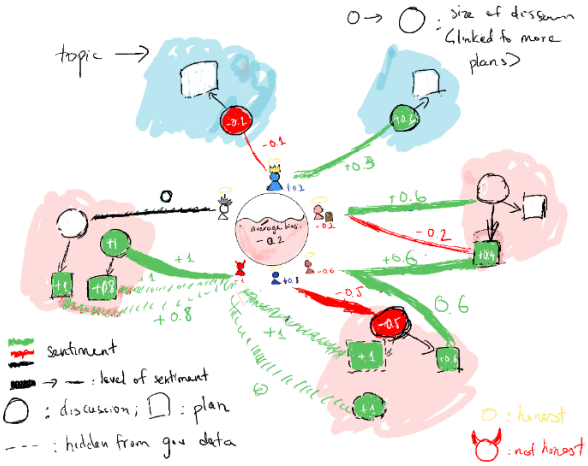
The sentiment biases chart aims to explain to the user the sentiment biases of the whole COOTEFOO board as well as each board member based on their participation tables from the journalist data. The table will show the average overall biases over all projects, as well as of each topic and each discussion or plan. The table will also shows the sentiments that are not disclosed on the government data and clearly indicate these differences.

Intended Design: based on cluster 1 of Converge

Display below is the hand-drawn visualization of the intended chart.

- In the center, there is a circle showing the board average sentiment as well as the human glyphs with icons showing their role / honesty:
  - a king's crown for the chairman, a queen's crown for the vice-chair, and a chest for the treasurer; others are normal members

- o the halo/devil-horn glyph indicates whether the person is publishing all of their sentiments in their government data or not
- From each person, there are connections linking each board member to each discussion/plan as well as their sentiment visualized by the color and thickness (pro-fishing is green, pro-tourism is red)
- Each group of topics is color coded by color, where there are also links from discussions to plans
- We also color-code the average sentiment for each discussion and plan. Additionally, to show concealed discussions and plans, the borders of these items, as well as any connections between them and the board members, are represented with dotted lines.



Actual Design

Intended versus actual design: While we are attempting to draw the visualization, there are some issues. First, there are too many sentiments linked between the board members and the plans. As such, if we attempted to draw them all, the visualization would be too messy if there is no easy way to easily move the groups around. Due to programming limitations, the locations of the topic groups are fixed. Furthermore, the linkage between discussions and plans may not be that necessary at all for the sentiment graph. Also, it's difficult to draw dynamic dotted lines for the link between the board member and the items. Nevertheless, the actual design mostly remains faithful to the intended design with some visual improvements for clarity.

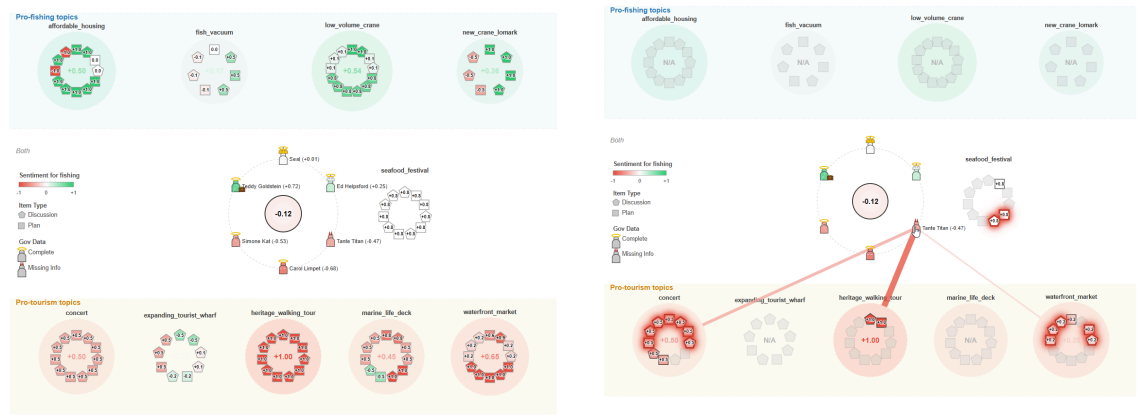


Figure: Actual implementaion cluster 1 (normal and when hovered)

The **visual elements** are:

- **Center board and human glyphs with roles and sentiment fill**: this is exactly the same as the intended design

- **Topics groups with discussions and plans:** this is also mostly the same as the intended design, but now the topics are clearly separated into focus (pro-fishing, pro-tourism or both) as well as the shapes are now colored coded to show the bias (more green is more pro-fishing, more red is pro-tourism)
- **Link from person to the topic:** instead of linking each person to each discussion/ plan, the link is now only to the topics, but the sentiment values will be updated on hover while blurring those that the person did not participate in.
- **Dishonesty glow:** there will be a glow in the discussion/ plans if the sentiments are only from the journalist's data.

The **interactions** are:

- **Hovering:** By default, the values and the color fill on the topics, sentiments, and plans will be the board average. When hovering over the specific board member, these will be shown as well as the linkage and dishonesty glow that they may have. For a clearer look, when hovering, the numbers and names for each of the non-hovered persons will be hidden and the activities that are not linked to the person will also be colored gray and made semi-transparent.
- **Legend:** Helps interpret the colors, shape and icons used in the graph, aiding quick understanding of the data.

Data used: Sentiment data (discussion_people_participations and plan_people_participations) of both Government and Journalist data. For this graph, all topics that are marked as other / vanity (e.g renaming places) are not included.

Link to code: https://github.com/tvakul/dataviz1/blob/main/index.py
Online link: https://tvakul.github.io/dataviz1

Implementation 2:

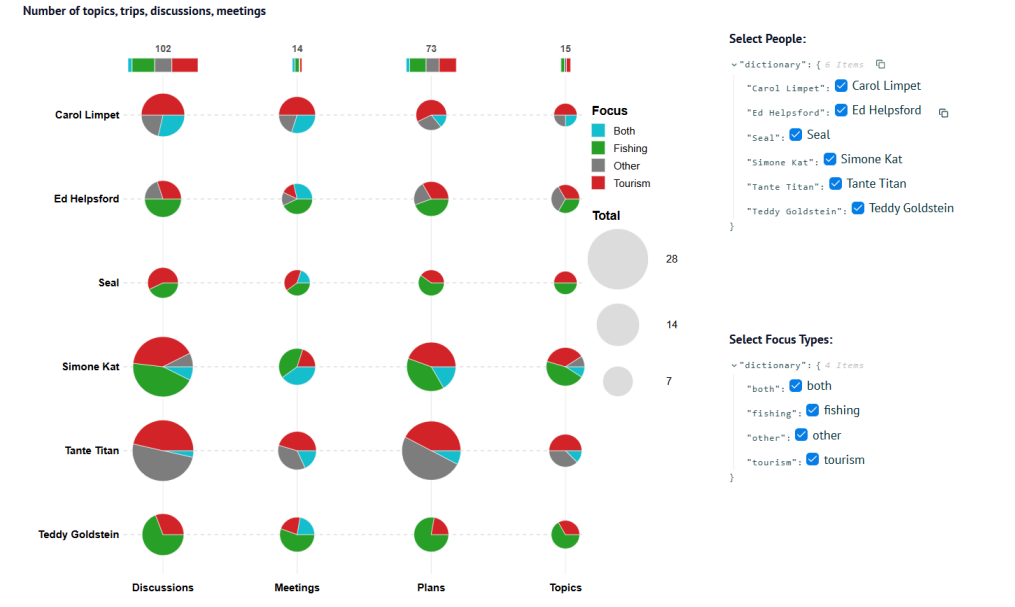
Implementation 3: Number of topics, plans, discussions, and meetings done by each member.

Description of the visual

This is a new design, which is loosely inspired by a graph design. This graph aims to show multiple things at the same time: who participates in most topics, plans, discussions or meetings, as well as the proportion of how many of them are related to pro-fishing or pro-tourism topics.

Intended Design:

This is a new design, which is loosely inspired by a graph design from the Diverge stage.



Actual Design

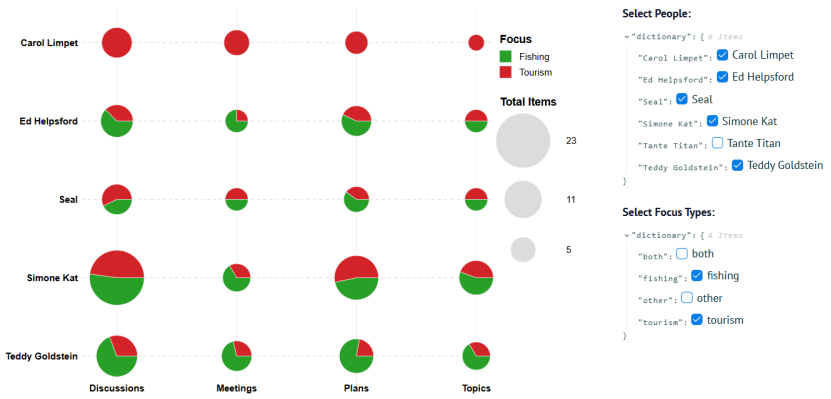
The implemented visualizations closely follow the intended design, incorporating dynamic and interactive features to enhance usability and insight generation.

The **visual elements are:**

- **Circle size:** represents the total number of participations of a board member of a certain type. On the vertical axis, serving as a position channel, showing all board members of COOTEFOO; on the other hand, the horizontal axis represents the participation types (Discussion, Meetings, Plans, and Topics). A larger circle means a larger number of participations.
- **Pie chart in the circle:** represents the proportion of each of the focus of the discussion/meetings/plans or topics. The larger the slice, the higher the proportion is.
- **Stacked vertical top bar** showing the total number of discussions/meetings/plans or topics by focus.
- **Legends** of the color of each of the focus types (green for pro-fishing, red for pro-tourism, blue for both, and gray for other) as well as a circle size reference for quick viewing.

The interactions are:

- **Hovering:** When hovering on each of the circles, the full number of each circle and its components will appear as a tooltip. For a visual hint, the circle hovered over is also highlighted.
- **Filters:** For more clarity, it is possible to filter out the person(s) or the focus group. When filtering, the total numbers (the circle size) as well as the proportions (pie slices) will be recalculated.



- **Legend:** Helps interpret the colors, shape and icons used in the graph, aiding quick understanding of the data.

Data used: Journalists' data of the board plans, discussions, topics and meetings

Link to code: https://github.com/tvakul/dataviz1/blob/main/index.py
Online link: https://tvakul.github.io/dataviz1

Part 5. Findings

In this part, describe any interesting/unexpected patterns or signals that you identified using your visualisations. Basically answer the question "what interesting things can you tell about the data"? You can use a story-telling approach: which initial patterns did you find when you started on the project using the question(s) that you defined in Part 2 point 3, how did that lead to focussing on specific parts in the data, etc.
For each finding, please include screenshots/images (if necessary/useful: annotated) that illustrate the patterns/insights that you found.

Max 2 pages.

Insights on sentiments / hidden sentiments (implementation 1)

- Overall, we can see that the average sentiment of the board is -0.12, meaning they are slightly biased against fishing. Furthermore, besides the mostly unbiased chairman, the vice-chair, and the persons that are pro-fishing, all other 3 board members are significantly anti-fishing and pro-tourism (their average sentiment scores are from -0.68 to -0.47). Topicwise, while the average sentiment of the board of pro-fishing topics is ranging from mixed (fish vacuum, 0.17) to moderately positive (low-volume crane at 0.36, affordable housing for fishermen at 0.5, and low-volume crane at 0.54), pro-fishing topics, except for the tourist wharf expansion, seem to be more preferable by the board (Marine life deck at 0.45, Music concert at 0.5, Waterfront Market at 0.65, and especially the Heritage Walking Tour has an unanimous score of 1.0).
- As for the three board members who are more negative against fishing, only Simone Kat seems to participate and have commented on her sentiment on the discussion or plans for both pro-fishing and pro-tourism topics. The two other people, Carol Limpet and Tante Titan, only have sentiments on pro-tourism topics or topics that benefit both tourism and fishing (the seafood festival), of which they all gave positive sentiment for.

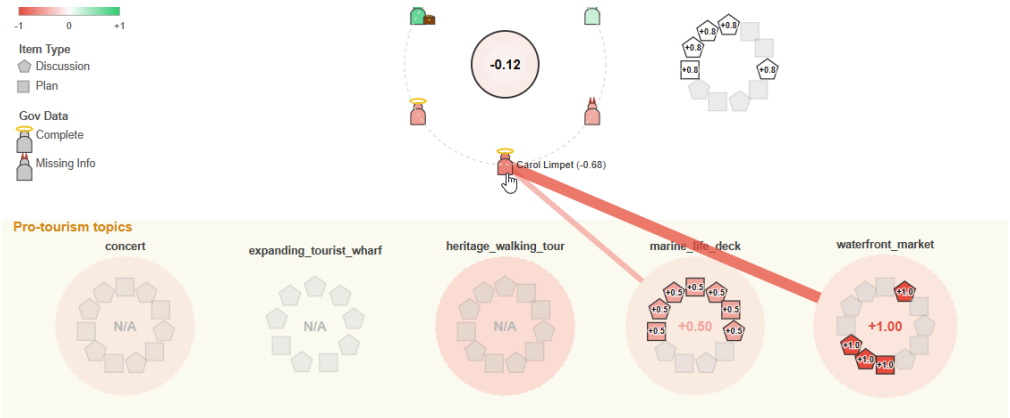


Figure: Sentiments of Carol Limpet

- Almost everyone on the boards has fully published their participations and sentiments on the government data, besides Tante Titan. Tante Titan only published 5 out of 20 sentiments on the government data. Almost all of the hidden sentiments are for the concerts and the waterfront market.

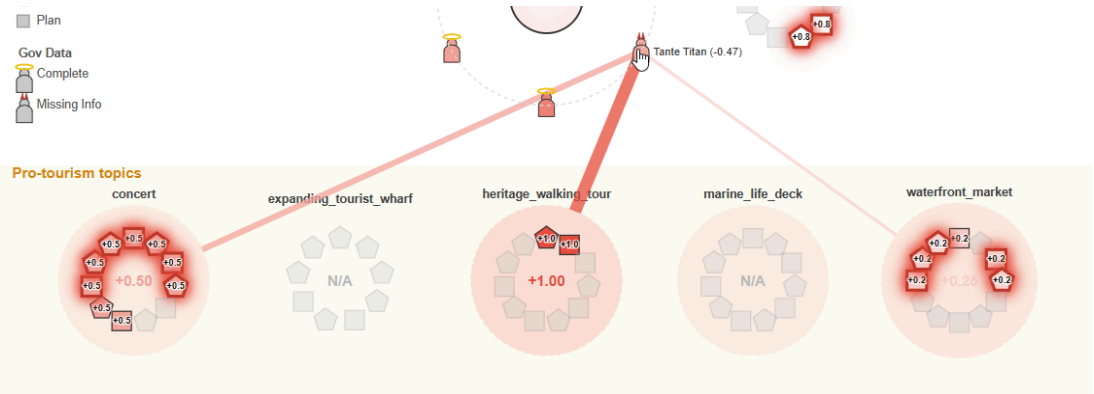
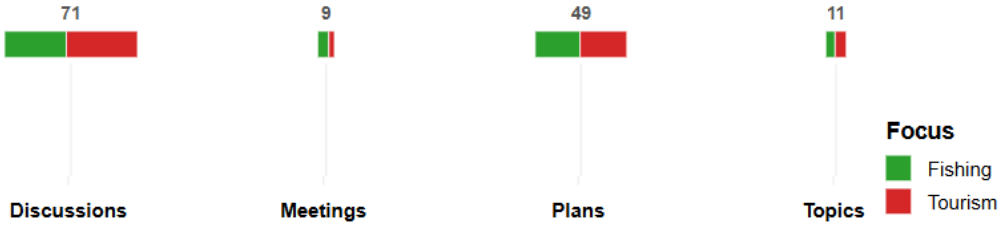


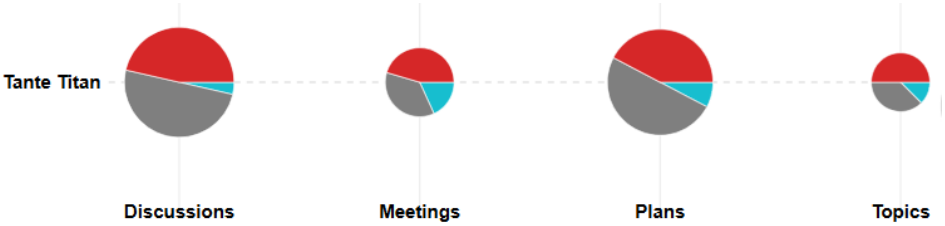
Figure: Sentiments of Tante Titan

Insights on the number of participations and proportions (implementation 3)

- From the top vertical bar charts, it can be seen that the number of total topics, discussions, and plans that benefit fishing is slightly smaller than the figure for tourism.



- It can be seen from the graph that the 3 normal board members Simone Kat, Tante Titan, and Carol Limpet participate in the most number of discussions, plans, and meetings. Nevertheless, it seems that Tante Titan seems to spend a majority of his effort on vanity projects, as the proportions of these topics, discussions, plans, and meetings of Tante are from 40% to over 50%.



- When filtering the focus to only fishing or tourism, it is clear that Tante and Carol participate in no meetings, plans, discussions, or topics related to pro-fishing. As such, maybe they should also participate in more pro-fishing projects.

Part 6. Reflections

This part will NOT be graded and optional: you do not need to answer these questions if you do not want to, or you could just answer one of them.
After the course you will be invited to a feedback query after the course so that we can improve for the future (keeping what is good, changing what could be improved). This part is therefore <u>not</u> about the course itself; it is about <u>your</u> own work, about what <u>you</u> learned or found: the way you collaborated, a specific design, finding out how to do something technically, etc.

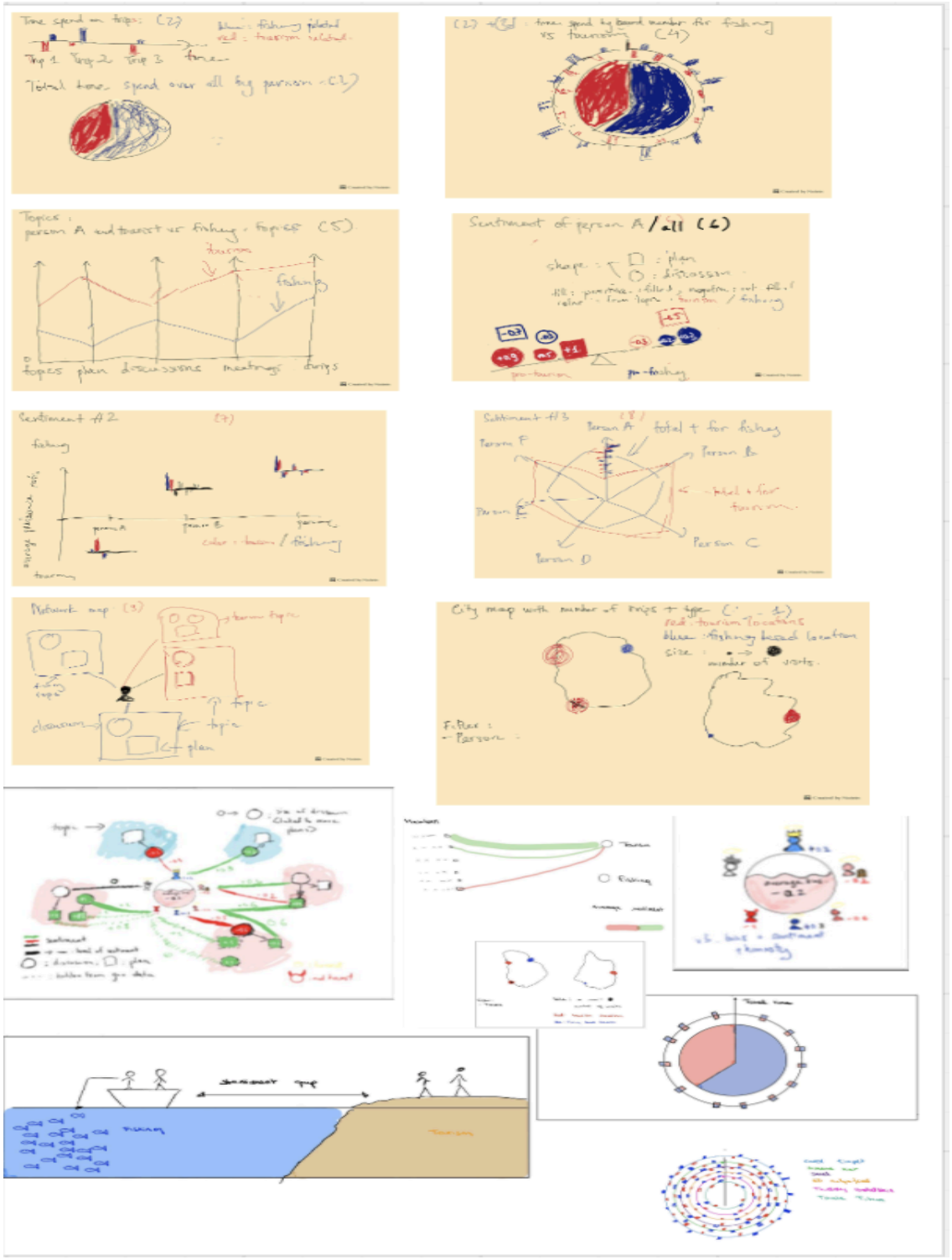
Specific questions that we would like an answer for.

1. *What are you <u>most proud</u> of in this project?*
2. *What are you <u>least proud</u> of in this project?*

Max 0.25 pages.

Part 7. Individual contributions

Describe the contribution of each of the different team members: who did what?
Max 0.25 pages.

Appendix A. Sketches overview

A single overview picture including all sketches that you created (so that the reviewers can assess the width of your exploration)
Max 1 page